Exp 16 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
Fs = 10000;
t = 0:1/Fs:0.02;
fm = 100;
n = 3;
L = 2^n;

% Message Signal
x = sin(2*pi*fm*t);

% Uniform Quantization
xmin = -1;
xmax = 1;
delta = (xmax-xmin)/L;
xq_uniform = xmin + delta/2 + delta*floor((x-xmin)/delta);
xq_uniform = min(max(xq_uniform,xmin+delta/2),xmax-delta/2);

% Non-Uniform Quantization using Mu-Law
mu = 255;
x_compressed = sign(x).*log(1+mu*abs(x))/log(1+mu);

delta2 = 2/L;
xq_compressed = -1 + delta2/2 + delta2*floor((x_compressed+1)/delta2);
```

```
xq_compressed = min(max(xq_compressed,-1+delta2/2),1-delta2/2);

xq_nonuniform = sign(xq_compressed).* ...
   ((1+mu).^abs(xq_compressed)-1)/mu;

% Calculate Quantization Errors
error_uniform = x-xq_uniform;
error_nonuniform = x-xq_nonuniform;

MSE_uniform = mean(error_uniform.^2);
MSE_nonuniform = mean(error_nonuniform.^2);

% Display Results
fprintf('Uniform Quantization MSE = %.6f\n',MSE_uniform);
fprintf('Non-Uniform Quantization MSE = %.6f\n',MSE_nonuniform);

% Plot Results
figure;

subplot(3,1,1);
plot(t,x);
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
stairs(t,xq_uniform);
```

```
title('Uniform Quantized Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

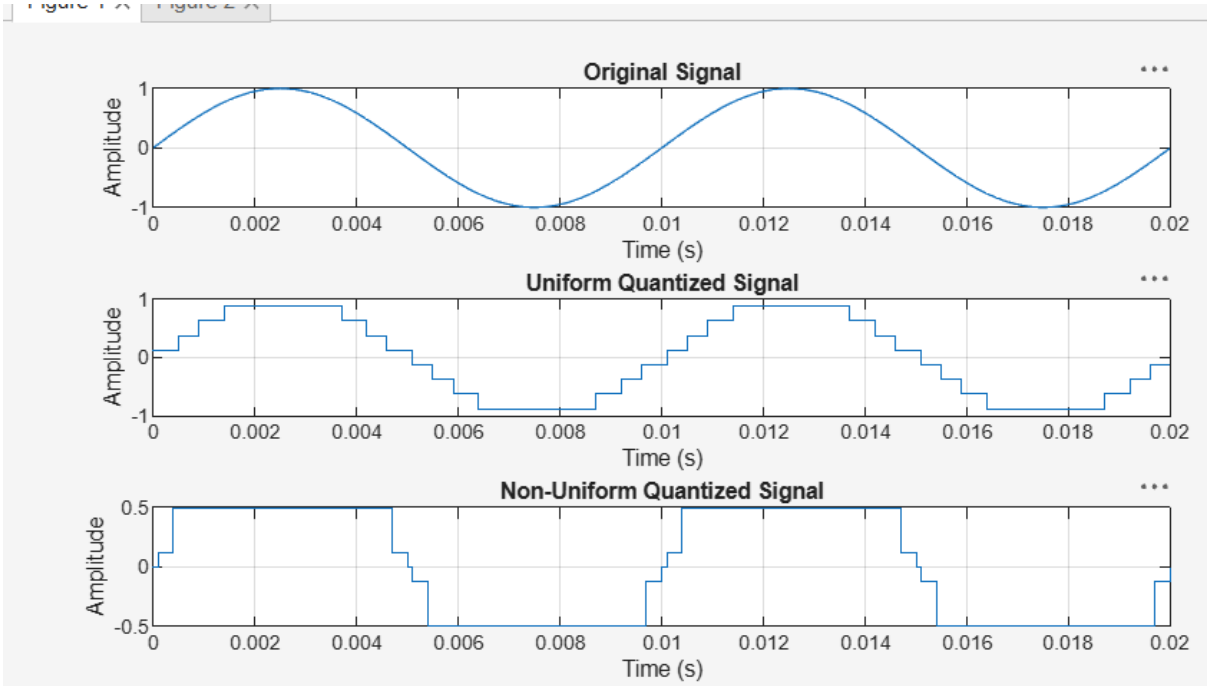
subplot(3,1,3);
stairs(t,xq_nonuniform);
title('Non-Uniform Quantized Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

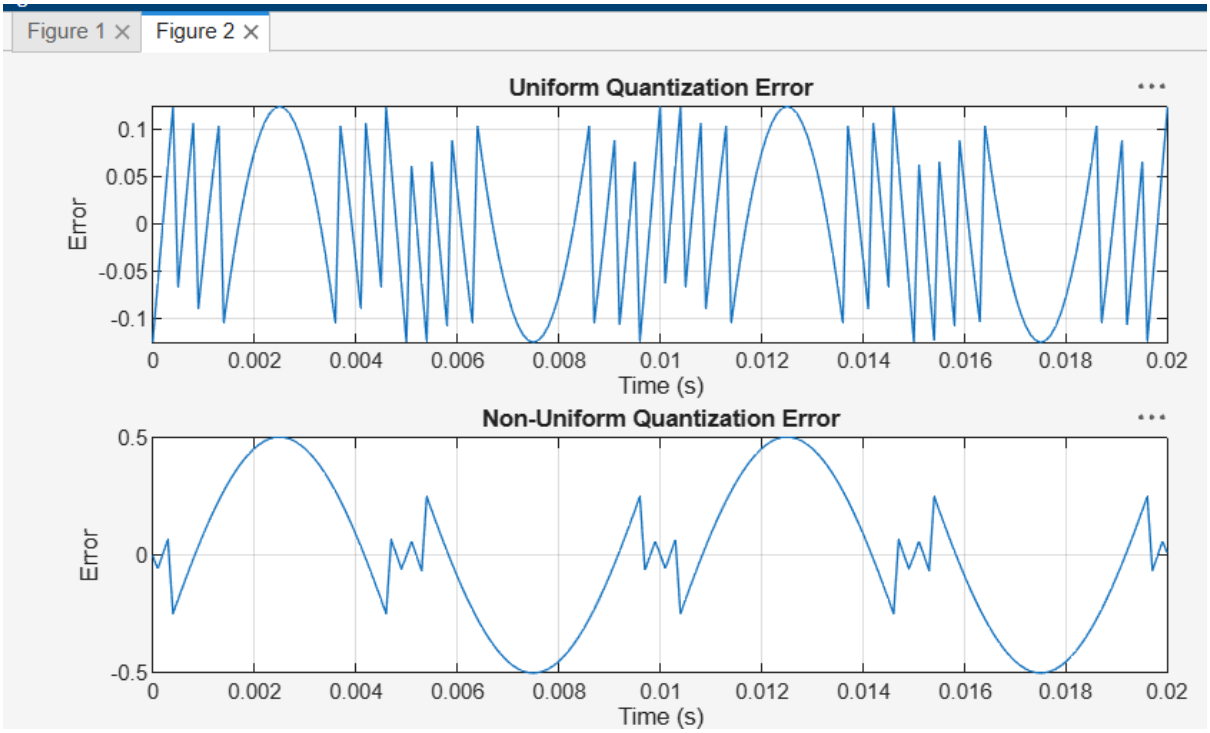
% Plot Quantization Errors
figure;

subplot(2,1,1);
plot(t,error_uniform);
title('Uniform Quantization Error');
xlabel('Time (s)');
ylabel('Error');
grid on;

subplot(2,1,2);
plot(t,error_nonuniform);
title('Non-Uniform Quantization Error');
xlabel('Time (s)');
ylabel('Error');
```

grid on;


Original Signal
Amplitude
Time (s)
Uniform Quantized Signal
Amplitude
Time (s)
Non-Uniform Quantized Signal
Amplitude
Time (s)


Figure 1
Figure 2
Uniform Quantization Error
Error
Time (s)
Non-Uniform Quantization Error
Error
Time (s)